

Human resources include the essential personnel routinely working with the pilot to ensure safety of flight. These people include, but are not limited to: weather briefers, flight line personnel, maintenance personnel, crew members, pilots, and air traffic personnel. Pilots need to communicate effectively with these people. This is accomplished by using the key components of the communication process: inquiry, advocacy, and assertion. Pilots should recognize the need to seek enough information from these resources to make a valid decision. After the necessary information has been gathered, the pilot's decision should be passed on to those concerned, such as air traffic controllers, crewmembers, and passengers. The pilot may have to request assistance from others and be assertive to resolve some situations safely.

Equipment in many of today's aircraft includes automated flight and navigation systems. These automatic systems, while providing relief from many routine tasks, present a different set of problems for pilots. The automation intended to reduce pilot workload essentially removes the pilot from the process of managing the aircraft, thereby reducing situational awareness and leading to complacency. Information from these systems needs to be continually monitored to ensure proper situational awareness. Pilots should be aware of both equipment capabilities and equipment limitations in order to manage those systems effectively and safely.

Information workloads and automated systems, such as autopilots, need to be properly managed to ensure a safe flight. By planning ahead, a pilot can effectively reduce workload during critical phases of flight and prevent erosion of performance. The pilot who effectively manages his or her workload completes routine tasks as early as possible to preclude the possibility of becoming overloaded and stressed in the later, more critical stages of the flight.

Task Management

Pilots have a limited capacity for information. Once information flow exceeds the pilot's ability to process the information mentally, any additional information becomes unattended or displaces other tasks and information already being processed. In addition, distraction and fixation impede the ability to process information. For example, if a pilot becomes distracted and fixates on an instrument light failure, the unnecessary focus displaces capability and prevents appreciation of tasks of greater importance.

Aeronautical Decision-Making (ADM)

Flying safely requires the effective integration of three separate sets of skills: stick-and-rudder skills needed to control the airplane; skills related to proficient operation of aircraft systems; and ADM skills. The ADM process addresses all aspects of decision-making in the flight deck and identifies the steps involved in good decision-making. While the ADM process does not eliminate errors, it helps the pilot recognize errors and enables the pilot to manage the error to minimize its effects. These steps are:

1. Identifying personal attitudes hazardous to safe flight;
2. Learning behavior modification techniques;
3. Learning how to recognize and cope with stress;
4. Developing risk assessment skills;
5. Using all resources; and
6. Evaluating the effectiveness of one's own personal ADM skills.

Ground Operations

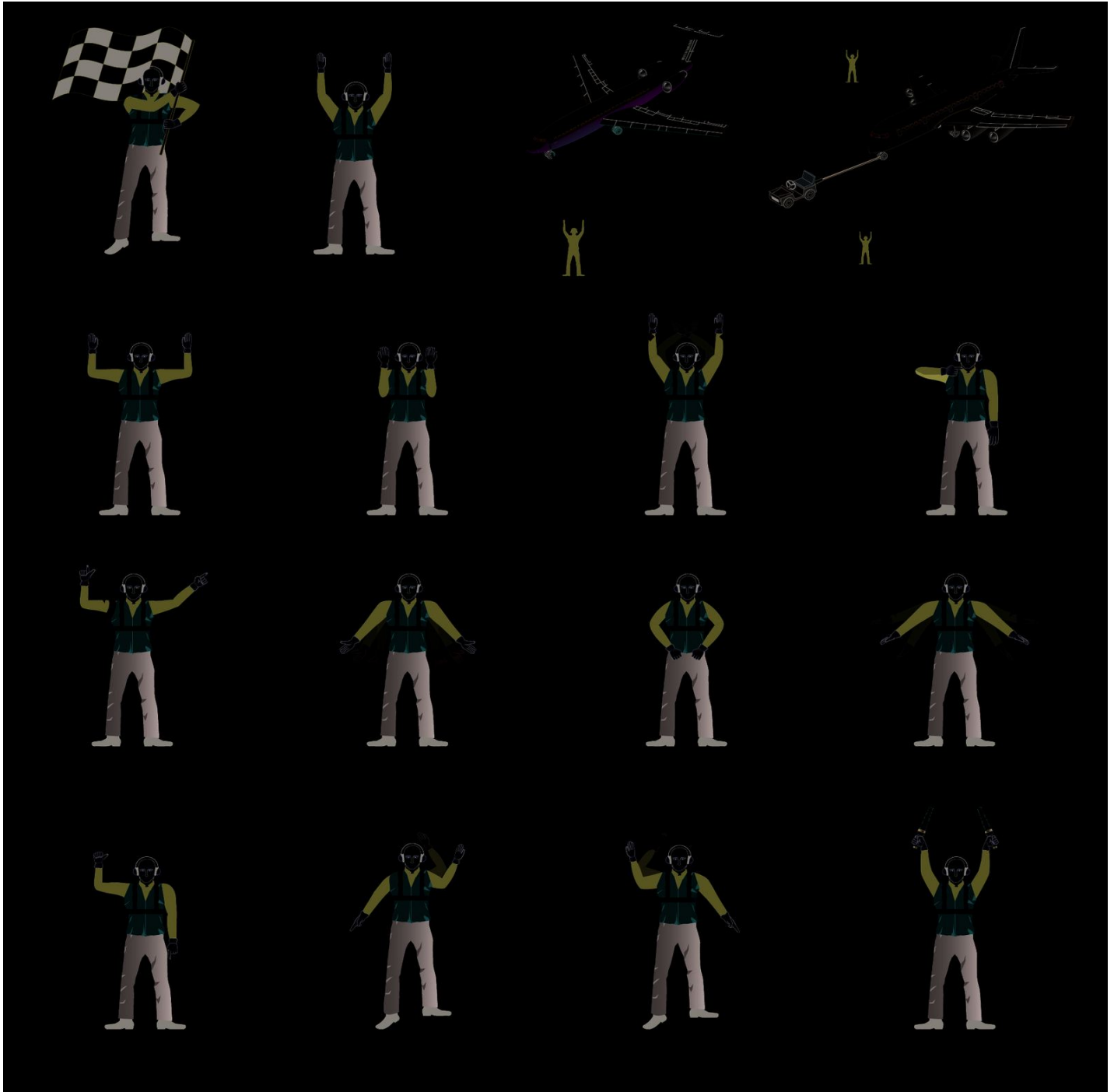
The airport ramp can be a complex environment with airport personnel, passengers, trucks, other vehicles, aircraft, and errant people and animals. The pilot is responsible for the operation of the airplane and should operate safely at all times. Ground operations subject the pilot to unique hazards, and mitigating those hazards requires proper planning and good situational awareness in the ground environment. A mitigation tactic involves reviewing the airport diagram prior to operating and having it readily available at all times. Whether departing to or from the ramp, the pilot needs to understand and capably manage the following:

1. Refueling operations
2. Passenger and baggage security and loading
3. Ramp and taxi operations
4. Standard ramp signals

During refueling operations, it is advisable that the pilot remove all passengers from the aircraft and witness the refueling to ensure that the correct fuel and quantity is dispensed into the airplane and that any caps and cowls are properly secured after refueling.

Passengers may have little experience with the open ramp of an airport. The pilot should ensure the safety of the passengers by cautioning them to move on the surface only as directed. If not under the pilot's direct supervision, passengers should have an escort to ensure their safety and ramp security. Baggage loading and security should also be supervised by the pilot. Unsecured baggage or improperly loaded baggage may adversely affect the center of gravity of the aircraft.

Some ramps may be staffed by personnel to assist the pilot in managing a safe departure from the ramp to the taxiway. *Figure 2-13* shows standard aircraft taxiing signals, such as those published in the Aeronautical Information Manual (AIM). There are other standard signals, such as those published in Advisory Circular 00-34, as revised, and by the Armed Forces. Furthermore, operation conditions in many areas may call for a modified set of taxi signals. The signals shown in *Figure 2-13* represent a minimum number of the most commonly used signals. Whether this set of signals or a modified set is used is not the most important consideration, as long as each flight operational center uses a suitable, agreed-upon set of signals.



2-15

Engine Starting

Airplane engines vary substantially and specific procedures for engine starting should be accomplished in reference to the approved engine start checklist as detailed in the airplane's AFM/POH. However, some generally accepted hazard mitigation practices and procedures are outlined in this section.

Prior to engine start, the pilot needs to ensure that the ramp area surrounding the airplane is clear of persons, equipment, and other hazards that could come into contact with the airplane or the propeller. Also, the pilot should check what is behind the airplane prior to engine start as standard practice. A propeller or other engine thrust can accelerate objects to substantial velocities, causing damage to property, and injuring those on the ground. The pilot should mitigate the hazard of debris being blown into persons or property. At all times before engine start, the anti-collision lights should be turned on. For night operations, the position (navigation) lights should also be on. Finally, just prior to starter engagement, the pilot should always call "CLEAR" out of the side window and wait for a response from anyone who may be nearby **before** engaging the starter.

When activating the starter, the wheel brakes need to be depressed and one hand kept on the throttle to manage the initial starting engine speed. Ensuring that properly operating brakes are engaged prior to starter engagement prevents the airplane from rapidly lunging forward. After engine start, the pilot manipulates the throttle to set the engine revolutions per minute (rpm) to the AFM/POH-prescribed setting. In general, 1,000 rpm is recommended following engine start to allow oil pressure to rise and to minimize undue engine wear due to insufficient lubrication at high rpm. It is important to service an airplane engine with the proper grade of oil for the seasonal conditions and to apply engine preheat when temperatures approach and descend below freezing.

The oil pressure should be monitored after engine start to ensure that pressure is increasing toward the AFM/POH-specified value. The AFM/POH specifies an oil pressure range for the engine. If the limits are not reached and maintained, serious internal engine damage is likely. In most conditions, oil pressure should rise to at least the lower limit within 30 seconds. To prevent damage, the engine should be shut down immediately if the oil pressure does not rise to the AFM/POH values within the required time.

Engine starters are electric motors designed to produce rapid rotation of the engine crankshaft for starting. These electric motors are **not** designed for continuous duty. Their service life may be drastically shortened during a prolonged or difficult start as an excess buildup of heat can damage internal starter components. Avoid continuous starter operation for periods longer than 30 seconds without a cool down period of at least 30 seconds to 1 minute (some AFM/POH specify longer cool down routines). The smell of burning insulation from a starter may indicate that the recommended cranking time has been exceeded. After repeated unsuccessful start attempts, the pilot should seek advice from a qualified person to determine the cause for the difficulty.

Although quite rare, the starter motor may remain electrically and mechanically engaged after engine start. This can be detected by a continuous and very high current draw on the ammeter. Some airplanes also have a starter engaged warning light specifically for this purpose. The engine should be shut down immediately if this occurs.

The pilot should be attentive for sounds, vibrations, smells, or smoke that are not consistent with normal after-start operational experience. Any concerns should lead to a shutdown and further investigation.

Hand Propping

The procedures for hand propping should always be in accordance with the AFM/POH and performed only by persons who are competent with hand propping procedures. The consequences of the hazards associated with hand propping are serious to fatal.

Historically, when aircraft lacked electrical systems, it was necessary for pilots and ground personnel to "hand prop" an aircraft for starting. Today, most airplanes are equipped with electric starters, and the starter should be working if the airplane is airworthy. If not, a certificated Aviation Maintenance Technician should be called to make a repair. However, vintage airplanes may be encountered, and an airplane manufactured without an electric starter needs to be hand propped. Since a number of these airplanes have been produced, the procedures for hand propping are described in this section.

A few simple precautions help to avoid accidents when hand propping the engine. While touching a propeller, always assume that the ignition is on. The switches that control the magnetos operate on the principle of short-circuiting the current to turn the ignition off. If the switch is faulty, it can be in the "off" position and still permit current to flow in the magneto primary circuit. This condition could allow the engine to start when the switch is off.

Hand propping an aircraft is a hazardous procedure when done perfectly. Not mitigating the hazards associated with hand propping can lead to serious injury and a runaway airplane. A spinning propeller can be lethal should it strike someone. Persons not trained, not competent, or who do not understand how to mitigate the hazards associated with hand propping should **never** perform this procedure!

Hand propping requires a team of two properly trained people. Both individuals should be familiar with the airplane and hand propping techniques. The first person is responsible for directing the procedure including pulling the propeller blades through. The second person sits in the airplane to ensure that the brakes are set and to exercise controls as directed by the person pulling the propeller. When hand propping occurs, a person unfamiliar with the controls should **never** occupy the pilot's seat.

When hand propping is necessary, the ground surface near the propeller should be stable and free of debris. Loose gravel, wet grass, grease, mud, oil, ice, or snow might cause the person pulling the propeller through to slip into the rotating blades as the engine starts. Unless a firm footing is available, relocate the airplane to mitigate this hazardous consequence.

Both participants should discuss the procedure and agree on voice commands and expected actions. To begin the procedure, the fuel system and engine controls (tank selector, primer, pump, throttle, and mixture) are set for normal start. The ignition/magneto switch should be checked to be sure that it is OFF. Then, the descending propeller blade should be rotated so that it assumes a position slightly above the horizontal. The person doing the hand propping should face the descending blade squarely and stand slightly less than one arm's length from the blade. If a stance too far away were assumed, it would be necessary to lean forward in an unbalanced condition to reach the blade, which may cause the person to fall forward into the rotating blades when the engine starts. Allowing space for the person to be able to step away as the propeller is pulled down, and the engine starts, serves as safeguard in case the brakes fail.

The procedure and commands for hand propping are:

- Person out front says, "FUEL ON, SWITCH OFF, THROTTLE CLOSED, BRAKES SET."
- Pilot seat occupant, after making sure the fuel is ON, mixture is RICH, magneto switch is OFF, throttle is CLOSED, and brakes are SET, says, "FUEL ON, SWITCH OFF, THROTTLE CLOSED, BRAKES SET."
- Person out front, after pulling the propeller through to prime the engine says, "BRAKES AND CONTACT."
- Pilot seat occupant checks the brakes SET and turns the magnetos switch ON, then says, "BRAKES AND CONTACT."

The words CONTACT (magnetos ON) and SWITCH OFF (magnetos OFF) are used because they are significantly different from each other. Under noisy conditions or high winds, the words CONTACT and SWITCH OFF are less likely to be misunderstood than SWITCH ON and SWITCH OFF.

The propeller is swung by forcing the blade downward rapidly, pushing with the palms of both hands. If the blade is gripped tightly with the fingers, the person's body may be drawn into the propeller blades should the engine misfire, "kickback," or rotate momentarily in the opposite direction. As the blade is pushed down, the person should step backward, away from the propeller. If the engine does not start, the propeller should not be repositioned for another attempt until it is verified that the magneto switch is turned OFF. Excessive throttle opening after the engine has fired is the principal cause of backfiring during starting. Gradual opening of the throttle, while the engine is cold, reduces the potential for backfiring. Slow, smooth movement of the throttle assures correct engine operation.

Immediately after the engine starts, check the oil pressure indicator. If oil pressure does not show within 30 seconds, stop the engine and determine the trouble. If oil pressure is indicated, adjust the throttle to the aircraft manufacturer's specified rpm for engine warmup, which is usually between 1,000 to 1,300 rpm.

Most aircraft reciprocating engines are air-cooled and depend on the forward speed of the aircraft to maintain proper cooling. Therefore, particular care is necessary when operating these engines on the ground. During all ground running, operate the engine with the propeller in full low pitch and headed into the wind with the cowlings installed to provide the best degree of engine cooling. Closely monitor the engine instruments at all times. Do not close the cowl flaps for engine warm-up, they need to be in the open position while operating on the ground. When warming up the engine, ensure that personnel, ground equipment that may be damaged, or other aircraft are not in the propeller wash.

When removing the wheel chocks or untying the tail after the engine starts, everyone involved should remember that the propeller is nearly invisible. Serious injuries and fatalities have occurred when people who have just started an engine walk or reach into the propeller arc to remove the chocks, reach the cabin, or when moving toward the tail of the airplane. Before the wheel chocks are removed, the throttle should be set to idle and the chocks approached only from the rear of the propeller. One should never approach the wheel chocks from the front or the side.

Taxiing

Taxiing is the controlled movement of the airplane under its own power while on the surface. Since an airplane is moved under its own power between a parking area and the runway, the pilot needs to understand and be proficient in taxi procedures.

A pilot should maintain situational awareness of the ramp, parking areas, taxiways, runway environment, and the persons, equipment and aircraft at all times. Without such awareness, safety may be compromised. Depending on the airport, the parking, ramp, and taxiways may or may not be controlled. As such, it is important that the pilot completely understands the operating environment. At small, rural airports these areas may be desolate with few aircraft and limited hazards; however, as the complexity of the airport increases so does the potential for hazards. Regardless of the complexity, some generally accepted procedures are appropriate.

- The pilot should be familiar with the parking, ramp, and taxi environment. This can be done by having an airport diagram, if available, out and in view at all times. [Figure 2-14]

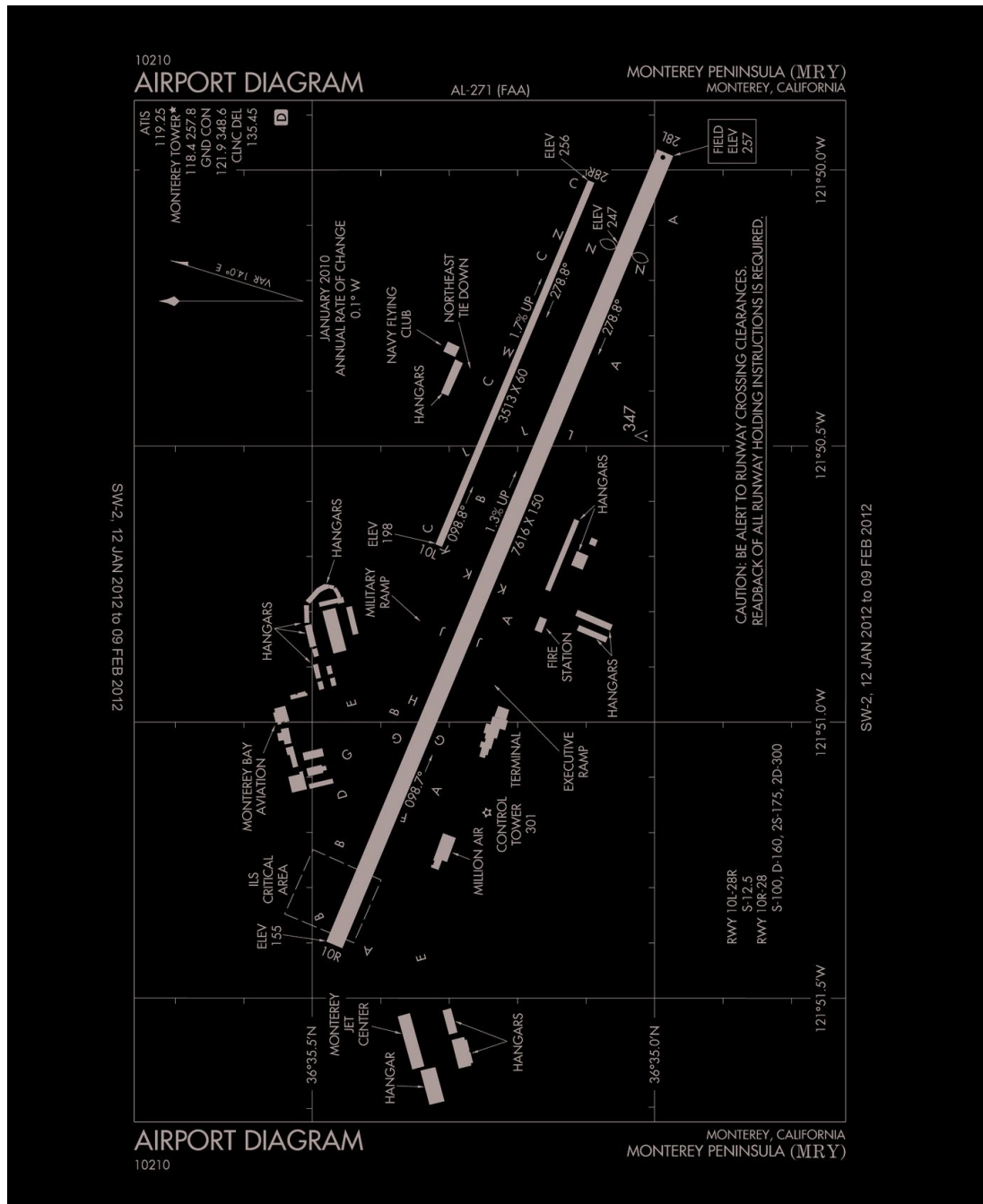


Figure 2-14. Airport Diagram of Monterey Peninsula (MRY), Monterey, California.

- Despite having familiarity with the airport, pilots should carefully review their complete taxi plan. For example, a pilot given the same taxi instructions by ATC, starts expecting those same instructions and might not realize that those instructions no longer apply. It only takes missing one instruction or turn to generate an accident. It is a human tendency to follow the same procedure over and over. This expectation bias has occurred to many pilots who did not stop and carefully consider and evaluate their taxi instructions.
- The pilot should be vigilant of the entire area around the airplane to ensure that the airplane clears all obstructions. If, at any time, there is doubt about a safe clearance from an object, the pilot should stop the airplane and check the clearance. It may be necessary to have the airplane towed or physically moved by a ground crew.
- When taxiing, the pilot's eyes should be looking outside the airplane scanning from side to side while looking both near and far to assess routing and potential conflicts.
- A safe taxiing speed should be maintained. The primary requirements for safe taxiing are positive control, the ability to recognize any potential hazards in time to avoid them, and the ability to stop or turn where and when desired, without undue reliance on the brakes. Pilots should proceed at a cautious speed on congested or busy ramps. Normally, the speed should be at the rate where movement of the airplane is dependent on the throttle. That is, slow enough so when the throttle is closed, the airplane can be stopped promptly.
- The pilot should place the aircraft on the taxiway center. Some taxiways have above-ground taxi lights and signage that could impact the airplane or propellers if the pilot does not exercise accurate control. When yellow taxiway centerline stripes are present, the pilot should visually place the centerline stripe so it is under the center of the airplane fuselage.
- When taxiing, the pilot should slow down before attempting a turn. Sharp high-speed turns place undesirable side loads on the landing gear and may result in tire damage or an uncontrollable swerve or a ground loop. Swerves are most likely to occur when turning from a downwind heading toward an upwind heading. In moderate to high-wind conditions, the airplane may weathervane increasing the swerving tendency.

Steering is accomplished with rudder pedals and brakes. To turn the airplane on the ground, the pilot should apply the rudder in the desired direction of turn and use the appropriate power or brake to control the taxi speed. The rudder pedal should be held in the direction of the turn until just short of the point where the turn is to be stopped. Rudder pressure is then released or opposite pressure is applied as needed.

More engine power may be required to start the airplane moving forward, or to start a turn, than is required to keep it moving in any given direction. When using additional power, the throttle should immediately be retarded once the airplane begins moving to prevent excessive acceleration.

The brakes should be tested for proper operation as soon as the airplane is put in motion. Applying power to start the airplane moving forward slowly, then retarding the throttle and simultaneously applying just enough pressure to one side, then the other to confirm proper function and reaction of both brakes. This is best if the airplane has individual left/right brakes to stop the airplane. If braking performance is unsatisfactory, the engine should be shut down immediately.

When taxiing at appropriate speeds in no-wind conditions, the aileron and elevator control surfaces have little or no effect on directional control of the airplane. These controls should not be considered steering devices and should be held in a neutral position.

When taxiing with a quartering headwind, the wing on the upwind side (the side that the wind is coming from) tends to be lifted by the wind unless the aileron control is held in that direction (upwind aileron UP). Moving the aileron into the UP position reduces the effect of the wind striking that wing, thus reducing the lifting action. This control movement also causes the downwind aileron to be placed in the DOWN position, thus a small amount of lift and drag on the downwind wing, further reducing the tendency of the upwind wing to rise. *[Figure 2-15]*

When taxiing with a quartering tailwind, the elevator should be held in the DOWN position, and the upwind aileron, DOWN. Since the wind is striking the airplane from behind, these control positions reduce the tendency of the wind to get under the tail and the wing and to nose the airplane over. The application of these crosswind taxi corrections helps to minimize the weathervaning tendency and ultimately results in easier steering. *[Figure 2-15]*

The presence of moderate to strong headwinds and/or a strong propeller slipstream creates lift on the horizontal tail surfaces and makes it necessary to control the pitch attitude while taxiing. The elevator control in nosewheel-type airplanes should be held in the neutral position, while in tailwheel-type airplanes, it should be held in the full aft position to hold the tail down unless the headwind gets very strong, which allows for an elevator position closer to neutral.

Downwind taxiing usually requires less engine power after the initial ground roll has begun, since the wind is pushing the airplane forward. To avoid overheating the brakes and controlling the airplane's speed when taxiing downwind, the pilot should keep engine power to a minimum. Rather than continuously riding the brakes to control speed, it is appropriate to apply brakes only occasionally. Other than sharp turns at low speed, the throttle should always be at idle before the brakes are applied. It is a common error to taxi with a power setting that requires controlling taxi speed with the brakes.

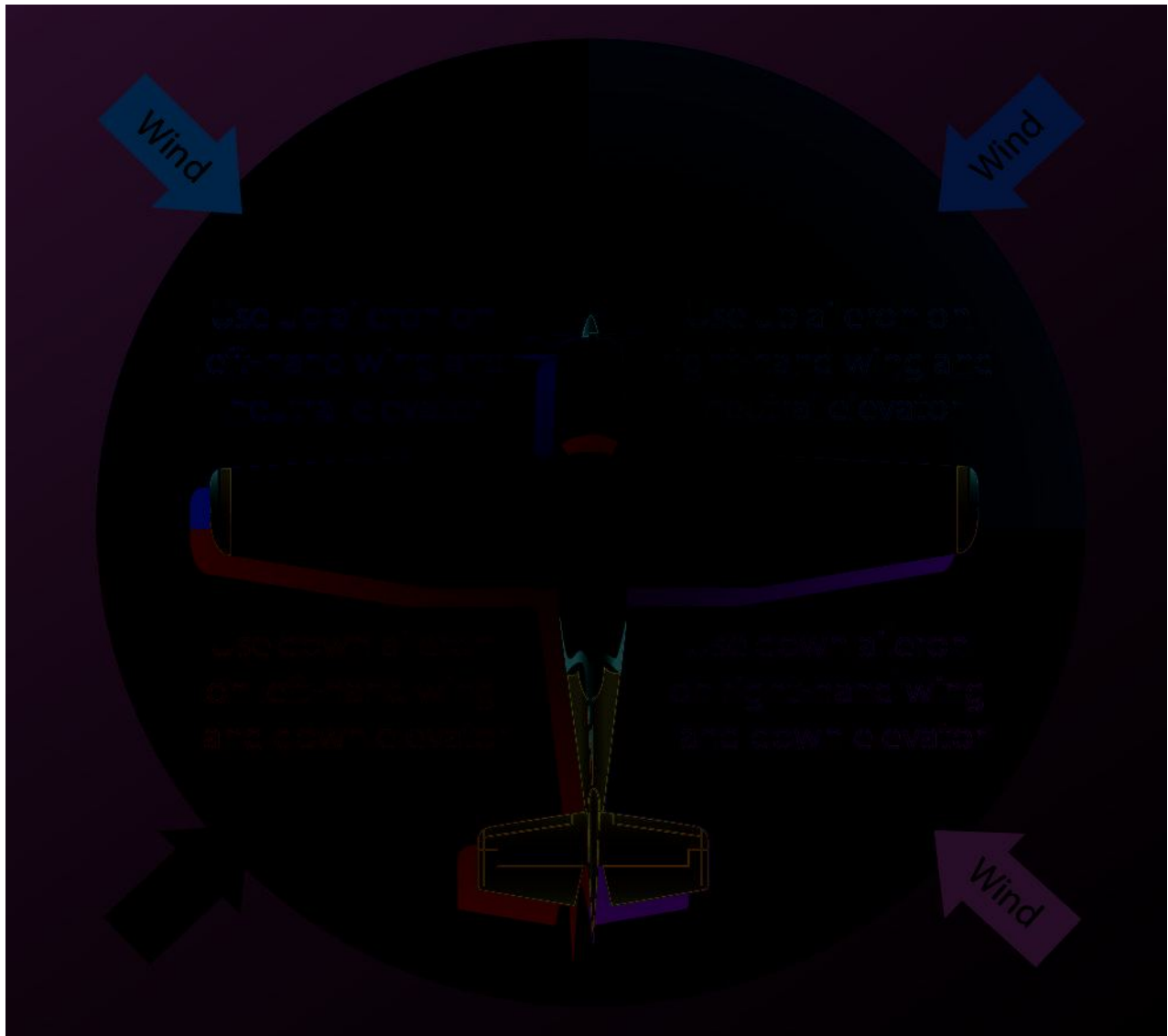


Figure 2-15. *Control positions of the nosewheel airplane.*

Normally, all turns should be started using the rudder pedal to steer the nosewheel. To tighten the turn after full pedal deflection is reached, the brake may be applied as needed. When stopping the airplane, it is always advisable to stop with the nosewheel straight ahead to relieve any side load on the nosewheel and to make it easier to start moving ahead. Note that certain makes and models have no nosewheel steering and the brakes need to be used to control any turns.

During crosswind taxiing, even the nosewheel-type airplane has some tendency to weathervane. However, the weathervaning tendency is less than in tailwheel-type airplanes because the main wheels are located behind the airplane's center of gravity, and the nosewheel's ground friction helps to resist the tendency. The nosewheel linkage from the rudder pedals provides adequate steering control for safe and efficient ground handling, and normally, only rudder pressure is necessary to correct for a crosswind.

Taxiing checklists are sometimes specified by the AFM/POH, and the pilot should accomplish any items that are required. If there are no specific checklist items, taxiing still provides an opportunity to verify the operation and cross-check of the flight instruments. In general, the flight instruments should indicate properly with the airspeed at or near zero (depending on taxi speed, wind speed and direction, and lower limit sensitivity); the attitude indicator should indicate pitch and roll level (depending on airplane attitude) with no flags; the altimeter should indicate the proper elevation within prescribed limits; the turn indicator should show the correct direction of turn with the ball movement toward the outside of the turn with no flags; the directional gyro should be set and crossed checked to the magnetic compass and verified accurate to the direction of taxi; and the vertical speed indicator (VSI) should read zero. These checks can be accomplished on conventional mechanical instrumented aircraft or those with glass displays.

Before-Takeoff Check

The before-takeoff check is the systematic AFM/POH procedure for checking the engine, controls, systems, instruments, and avionics prior to flight. Normally, the before-takeoff checklist is performed after taxiing to a run-up position near the takeoff end of the runway. Many engines require that the oil temperature reach a minimum value as stated in the AFM/POH before takeoff power is applied. Taxiing to the run-up position usually allows sufficient time for the engine to warm up to at least minimum operating temperature; however, the pilot should verify that the oil temperature is within the proper range prior to the application of high power.

A suitable location for run-up should be firm (a smooth, paved or turf surface if possible) and free of debris. Otherwise, the propeller may pick up pebbles, dirt, mud, sand, or other loose objects and hurl them backwards. This damages the propeller and may damage the tail of the airplane. Small chips in the leading edge of the propeller form stress risers or high stress concentrations. These are highly undesirable and may lead to cracks and possible propeller blade failure. The airplane should also be positioned clear of other aircraft and the taxiway. There should not be anything behind the airplane that might be damaged by the propeller airflow blasting rearward.

Before beginning the before-takeoff check, after the airplane is properly positioned for the run-up, it should be allowed to roll forward slightly to ensure that the nosewheel or tailwheel is in alignment with the longitudinal axis of the airplane.

While performing the before-takeoff check in accordance with the airplane's AFM/POH, the pilot divides attention between the inside and outside of the airplane. If the parking brake slips, or if application of the toe brakes is inadequate for the amount of power applied, the airplane could rapidly move forward and go unnoticed if pilot attention is fixed only inside the airplane. A good operational practice is to split attention from one item inside to a look outside.

Air-cooled engines generally are tightly cowled and equipped with baffles that direct the flow of air to the engine in sufficient volumes for cooling while in flight; however, on the ground, much less air is forced through the cowling and around the baffling. Prolonged ground operations may cause cylinder overheating long before there is an indication of rising oil temperature. To minimize overheating during engine run-up, it is recommended that the airplane be headed as nearly as possible into the wind and, if equipped, engine instruments that indicate cylinder head temperatures should be monitored. Cowl flaps, if available, should be set according to the AFM/POH.

Each airplane has different features and equipment and the before-takeoff checklist provided in airplane's AFM/POH should be used to perform the run-up. Many critical systems are checked and set during the before-takeoff check. Most airplanes have at least the following systems checked and set:

- Fuel System—set per the AFM/POH and verified ON and the proper and correct fuel tanks selected.
- Trim—set for takeoff position, which includes the elevator and may also include rudder and aileron trim.
- Flight Controls—checked throughout their entire operating range. This includes full aileron, elevator, and rudder deflection in all directions. Often, pilots do not exercise a full range of movement of the flight controls, which is not acceptable.
- Engine Operation—checked to ensure that temperatures and pressures are in their normal ranges; magneto or Full Authority Digital Engine Control (FADEC) operation on single or dual ignition are acceptable and within limits; and, if equipped, carburetor heat is functioning. If the airplane is equipped with a constant speed or feathering propeller, that its operation is acceptable, and the engine continues to run normally as the propeller is exercised.
- Electrical System—verified to ensure voltages are within operating range and that the system shows the battery system charging.

- Vacuum System—shows an acceptable level of vacuum, which is typically between 4.8 and 5.2 inches of mercury ("Hg) at 2,000 rpm. Refer to the AFM/POH for the manufacturer's values. It is important to ensure that mechanical gyroscopic instruments have adequate time to spool up to acceptable rpm in order for them to indicate properly. A hasty and quick taxi and run-up does not allow mechanical gyroscopic instruments to indicate properly and a departure into instrument meteorological conditions (IMC) is inadvisable.
- Flight Instruments—rechecked and set for the departure. Verify that the directional gyro and the magnetic compass are in agreement. If the directional gyro has a heading bug, it may be set to the runway heading that is in use or as assigned by air traffic control (ATC).
- Avionics—set with the appropriate frequencies, initial navigation sources and courses, autopilot preselects, transponder codes, and other settings and configurations based on the airplane's equipment and flight requirements.
- Takeoff Briefing—made out loud by the pilot even when no other person is there to listen. It should include a visual verification of the correct surface and direction to preclude a wrong surface departure. A sample takeoff briefing may be the following:

“This will be normal takeoff (use normal, short, or soft as appropriate) from runway (use runway assigned), wind is from the (direction and speed), rotation speed is (use the specified or calculated manufacturer's takeoff or rotation speed (V_R)), an initial turn to (use planned heading) and climb to (use initial altitude in feet). The takeoff will be rejected for engine failure below V_R , applying appropriate braking, stopping ahead. Engine failure after V_R and with runway remaining, I will lower pitch, land, and apply appropriate braking, stopping straight ahead. Engine failure after V_R and with no runway remaining, I will lower pitch to best glide speed, no turns will be made prior to (insert appropriate altitude), land in the most suitable area, and apply appropriate braking, avoiding hazards on the ground as much as possible. I will only consider turning back to runway ___ if I have reached at least ___ feet AGL, which would be ___ feet MSL. If time permits, fuel, ignition, and electrical systems will be switched off.”

Takeoff Checks

The pilot should ensure that runway numbers on paved runways agree with magnetic compass and heading indicators before beginning takeoff roll. The last check as power is brought to full takeoff power includes:

1. Doors latched and windows closed as required?
2. Controls positioned to account for any crosswind?
3. Power correct?
4. Engine rpm normal?
5. Engine smooth?
6. Engine instruments normal and in green ranges?

After-Landing

During the after-landing roll, while maintaining airplane track over runway centerline with ailerons and heading down runway with rudder pedals, the airplane should be gradually slowed to normal taxi speed with normal brake pressure before turning off of the landing runway. Any significant degree of turn at faster speeds could result in subsequent damage to the landing gear, tires, brakes, or the airplane structure.

To give full attention to controlling the airplane during the landing roll, the after-landing checklist should be performed only after the airplane is brought to a complete stop beyond the runway holding position markings. There have been many cases where a pilot has mistakenly manipulated the wrong handle and retracted the landing gear, instead of the flaps, due to improper division of attention while the airplane was moving. However, this procedure may be modified if the manufacturer recommends that specific after-landing items be accomplished during landing rollout. For example, when performing a short-field landing, the manufacturer may recommend retracting the flaps on rollout to improve braking. In this situation, the pilot should make a positive identification of the flap control handle before retracting the flaps.

Clear of Runway and Stopped

Because of different configurations and equipment in various airplanes, the after-landing checklist within the AFM/POH should be used. Some of the items may include:

1. Power—set to the AFM/POH values such as throttle 1,000 rpm, propeller full forward, mixture leaned.
2. Fuel—may require switching tanks and fuel pumps switched off.
3. Flaps—set to the retracted position.
4. Cowl flaps—may be opened or closed depending on temperature conditions.
5. Trim—reset to neutral or takeoff position.
6. Lights—may be switched off if not needed, such as strobe lights.
7. Avionics—frequencies and transponder set for arrival airport taxi procedures.

Parking

Unless parking in a designated, supervised area, the pilot should select a location and heading that prevents propeller or jet blast of other airplanes from striking the airplane unnecessarily. Whenever possible, the airplane should be parked headed into the existing or forecast wind. Often airports have airplane tie downs located on ramp areas which may or may not be aligned with the wind or provide a significant choice in parking location. After stopping in the desired direction, the airplane should be allowed to roll straight ahead enough to straighten the nosewheel or tailwheel.

Engine Shutdown

The pilot should always use the procedures in the airplane's AFM/POH shutdown checklist for shutting down the engine and securing the airplane. Important items may include:

1. Parking Brake—set to ON.
2. Throttle—set to IDLE or 1,000 rpm.
3. If turbocharged, observe the manufacturer's spool down procedure.
4. Magneto Switch Test—momentarily check for proper grounding in the OFF position at idle rpm.
5. Propeller Control—set to HIGH rpm, if equipped.
6. Avionics—turn OFF.
7. Alternator—turn OFF.
8. Mixture—set to IDLE CUTOFF.
9. Magneto Switch—turn ignition switch to OFF when engine stops.
10. Install chocks (release parking brake in accordance with AFM/POH).
11. Master Switch—turn OFF.
12. Secure—install control locks and anti-theft security locks.

Post-Flight

A flight is not complete until the engine is shut down and the airplane is secured. A pilot should consider this an essential part of any flight.

Securing and Servicing

After engine shutdown and deplaning passengers, the pilot should accomplish a post-flight inspection. This includes a walk around to inspect the general condition of the aircraft. Inspect near and around the cowling for signs of oil or fuel streaks and around the oil breather for excessive oil discharge. Inspect under wings and other fuel tank locations for fuel stains. Inspect landing gear and tires for damage and brakes for any leaking hydraulic fluid. Inspect cowling inlets for obstructions.

Oil levels should be checked and quantities brought to AFM/POH levels. Fuel should be added based on the immediate use of the airplane. If the airplane is going to be inactive, it is a good operating practice to fill the fuel tanks to prevent water condensation from forming inside the tank. If another flight is planned, the fuel tanks should be filled based on the flight planning requirements for that flight.

The aircraft should be hangared or tied down, flight controls secured, and security locks in place. The type of tie downs may vary significantly from chains to well-worn ropes. Chains are not flexible and as such should not be made taut so as to allow the airplane some movement and prevent airframe structural damage. Tie down ropes are flexible and may be reasonably cinched to the airplane's tie down rings. Consider utilizing pitot tube covers, cowling inlet covers, rudder gust locks, window sunscreens, and propeller security locks to further enhance the safety and security of the airplane.

Hangaring is not without hazards to the airplane. The pilot should ensure that enough space is allocated to the airplane so it is free from any impact to the hangar, another aircraft, or vehicle. The airplane should be inspected after hangaring to ensure that no damage was imparted on the airplane.